

Practical No. 12

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Aim : Identify risks involved in the project and prepare RMMM (Risk Management, Mitigation and Monitoring) Plan

Aim

To identify the possible risks involved in the **Online Food Ordering System** and prepare a **Risk Management, Mitigation and Monitoring (RMMM) Plan**.

Project Description

The Online Food Ordering System allows customers to browse restaurants and food items, add items to the cart, place orders, make online payments, and track their orders. Restaurant staff can manage food items and orders, while the administrator can manage users, restaurants, and system data.

1. Risk Identification

A **risk** is an uncertain event or condition that may negatively affect the project's cost, schedule, quality, security, or successful completion.

Sr. No.	Risk	Category	Probability	Impact
1	Requirement changes during development	Project	Medium	High
2	Delay in project development	Schedule	Medium	High
3	Database failure or data loss	Technical	Low	High
4	Server or hosting failure	Technical	Medium	High
5	Payment transaction failure	Technical	Medium	High
6	Unauthorized access to user accounts	Security	Medium	High
7	Leakage of customer information	Security	Low	High
8	Internet connectivity problems	External	Medium	Medium
9	Bugs in the application	Technical	High	Medium
10	Lack of team member availability	Project	Medium	Medium
11	Incorrect order information	Operational	Medium	High
12	Third-party API/service failure	External	Low	High

2. RMMM Plan

RMMM stands for **Risk Management, Risk Mitigation and Risk Monitoring.**

Sr.	Risk	Risk Management	Mitigation Plan	Monitoring Plan
1	Requirement changes	Discuss requirements with client before development.	Finalize and document requirements.	Review requirements during each development phase.
2	Development delay	Prepare a project schedule.	Divide work into smaller tasks and set deadlines.	Check task completion regularly.
3	Database failure/data loss	Identify important database information.	Take regular database backups.	Check backup status periodically.
4	Server/hosting failure	Select a reliable hosting service.	Maintain backups and recovery procedures.	Monitor server uptime and errors.
5	Payment failure	Identify possible payment-related issues.	Provide transaction status and retry options.	Monitor failed and successful transactions.
6	Unauthorized account access	Identify security vulnerabilities.	Use strong passwords, authentication and access control.	Review login attempts and security logs.
7	Customer data leakage	Identify sensitive information.	Use encryption and restrict database access.	Check access logs and security alerts.
8	Internet connectivity problems	Consider dependency on internet connection.	Display suitable error messages and allow retry.	Monitor connection and request failures.
9	Application bugs	Perform testing throughout development.	Conduct unit, integration and system testing.	Track and fix reported bugs.
10	Team member unavailable	Identify critical responsibilities.	Assign backup team members and maintain documentation.	Review team availability regularly.
11	Incorrect order information	Identify important order fields.	Validate order details before confirmation.	Check order records and customer complaints.
12	Third-party API failure	Identify external service dependencies.	Provide fallback/retry mechanisms where possible.	Monitor API response and failure rates.

3. Risk Priority

Risk Level	Meaning	Action
High	High probability and/or high impact	Immediate attention required
Medium	Moderate probability or impact	Regular monitoring required
Low	Low probability and impact	Monitor periodically

Major Risks Requiring Special Attention

1. Data loss or database failure
2. Payment transaction failure
3. Unauthorized access
4. Customer data leakage
5. Server failure
6. Application bugs
7. Incorrect order information
8. Requirement changes

4. Risk Monitoring

- Review project progress regularly.
- Check whether development tasks are completed on time.
- Test the application after every major feature.
- Monitor server and database performance.
- Maintain regular database backups.

- Check failed payment transactions.
- Review application and security logs.
- Record and resolve reported bugs.
- Review changes in project requirements.
- Update the RMMM plan whenever a new risk is identified.

5. Conclusion

The possible risks associated with the **Online Food Ordering System** were identified and analyzed. An **RMMM Plan** was prepared to define the management, mitigation, and monitoring activities for each risk. Proper risk management helps reduce project failure, improve software quality, maintain security, and complete the project successfully.

Result

The risks involved in the Online Food Ordering System were successfully identified and an RMMM (Risk Management, Mitigation and Monitoring) Plan was prepared.